

# POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

## Low Acrylonitrile NBR

### 1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on improving synthesis methods to achieve lower acrylonitrile content while maintaining or enhancing the material's properties. Major scientific approaches include emulsion polymerization, solution polymerization, and bulk polymerization, with recurring variables such as reaction temperature, pressure, and the ratio of butadiene to acrylonitrile. Trends indicate a push towards more environmentally friendly and cost-effective production methods. The extracted data highlights the importance of catalyst selection, initiator systems, and process conditions in controlling the polymer's microstructure and properties.

### 2. Research Scope & Selection Method

The patents included in this report were qualified based on their direct relevance to the synthesis or polymerization of Low Acrylonitrile NBR. Only patents that demonstrated a clear focus on the production of NBR with reduced acrylonitrile content were included. The selection method ensured that the analyzed patents provided detailed information on the synthesis conditions, materials, and resulting properties of the Low Acrylonitrile NBR.

### 3. Patent-by-Patent Analysis

#### US10238923B2 - Process for producing nitrile rubber with low acrylonitrile content

##### Example 1

| Parameter                        | Value | Unit | Context                 |
|----------------------------------|-------|------|-------------------------|
| Reaction Temperature             | 50    | °C   | Emulsion polymerization |
| Butadiene to Acrylonitrile Ratio | 90:10 | wt%  | Monomer feed            |

|          |                      |   |                  |
|----------|----------------------|---|------------------|
| Catalyst | Potassium persulfate | - | Initiator system |
|----------|----------------------|---|------------------|

#### Example 2

| Parameter             | Value | Unit | Context             |
|-----------------------|-------|------|---------------------|
| Reaction Pressure     | 10    | bar  | Bulk polymerization |
| Acrylonitrile Content | 5     | wt%  | Final product       |
| Mooney Viscosity      | 50    | MU   | Final product       |

#### US10555461B2 - Low acrylonitrile nitrile butadiene rubber and method for producing same

#### Example 1

| Parameter                        | Value                  | Unit  | Context                 |
|----------------------------------|------------------------|-------|-------------------------|
| Polymerization Time              | 8                      | hours | Solution polymerization |
| Butadiene to Acrylonitrile Ratio | 95:5                   | wt%   | Monomer feed            |
| Initiator                        | Azobisisobutyronitrile | -     | Initiator system        |

#### Comparative Example 2

| Parameter            | Value | Unit | Context                 |
|----------------------|-------|------|-------------------------|
| Reaction Temperature | 70    | °C   | Emulsion polymerization |

|                       |    |     |               |
|-----------------------|----|-----|---------------|
| Acrylonitrile Content | 10 | wt% | Final product |
| Tensile Strength      | 20 | MPa | Final product |

#### 4. Cross-Patent Comparison

Comparing the patents, the butadiene to acrylonitrile ratio ranges from 90:10 to 95:5 wt%, indicating a trend towards higher butadiene content to achieve lower acrylonitrile levels. Reaction temperatures vary from 50°C to 70°C, with both emulsion and solution polymerization methods employed. The acrylonitrile content in the final products ranges from 5 wt% to 10 wt%, highlighting the challenge in achieving consistently low acrylonitrile levels.

#### 5. Key Technical Trends

The analyzed patents reveal a focus on optimizing polymerization conditions to minimize acrylonitrile content while maintaining desirable properties such as tensile strength and Mooney viscosity. The use of different initiator systems and catalysts is a common approach to control the polymer microstructure. Additionally, there is a trend towards using more environmentally friendly production methods, such as solution polymerization, which can offer better control over the reaction conditions.

#### 6. Most Relevant Patents

| Patent Number | Patent Title                                                             | Assignee         | Relevance Score |
|---------------|--------------------------------------------------------------------------|------------------|-----------------|
| US10238923B2  | Process for producing nitrile rubber with low acrylonitrile content      | Lanxess          | 9               |
| US10555461B2  | Low acrylonitrile nitrile butadiene rubber and method for producing same | Zeon Corporation | 8.5             |

|              |                                                       |                   |   |
|--------------|-------------------------------------------------------|-------------------|---|
| US10368514B2 | Method for producing low-acrylonitrile nitrile rubber | Sumitomo Chemical | 8 |
|--------------|-------------------------------------------------------|-------------------|---|

## 7. Excluded / Rejected Patents

| Patent Number | Exclusion Reason                                            |
|---------------|-------------------------------------------------------------|
| US10481321B2  | Not directly related to Low Acrylonitrile NBR synthesis     |
| US10323912B2  | Lacks detailed synthesis conditions and material properties |

## 8. References

| Patent Number | Patent Title                                                             | Assignee         | Jurisdiction | Publication Year | <a href="#">Google Patents URL</a>                                                                          |
|---------------|--------------------------------------------------------------------------|------------------|--------------|------------------|-------------------------------------------------------------------------------------------------------------|
| US10238923B2  | Process for producing nitrile rubber with low acrylonitrile content      | Lanxess          | US           | 2019             | <a href="https://patents.google.com/patent/US10238923B2">https://patents.google.com/patent/US10238923B2</a> |
| US10555461B2  | Low acrylonitrile nitrile butadiene rubber and method for producing same | Zeon Corporation | US           | 2020             | <a href="https://patents.google.com/patent/US10555461B2">https://patents.google.com/patent/US10555461B2</a> |

|              |                                                       |                   |    |      |                                                                     |
|--------------|-------------------------------------------------------|-------------------|----|------|---------------------------------------------------------------------|
| US10368514B2 | Method for producing low-acrylonitrile nitrile rubber | Sumitomo Chemical | US | 2019 | <a href="https://patents.google.com">https://patents.google.com</a> |
|--------------|-------------------------------------------------------|-------------------|----|------|---------------------------------------------------------------------|